Article II: Technology

Technology is an essential part of the conversation when it comes to the direction of the future. The establishment of the Internet and the simultaneous rapid advancement of technology has made the world a totally different place than it was in previous decades–and that advancement is exponentially increasing.

The climate crisis is a broad, layered issue, and technology is one of those layers that contribute immensely to environmental degradation. Cryptocurrency and data centers are two niches of the tech sector that are responsible for some of the most energy use. For example, Bitcoin alone is responsible for roughly 150 terawatt-hours of annual electricity use, and "producing that energy emits some 65 megatons of carbon dioxide into the atmosphere annually" (Hinsdale). That much carbon dioxide being released into the atmosphere is incredibly harmful because it pollutes the air, especially considering how many other sources of carbon dioxide exist.

Data centers also require a massive amount of energy to operate. A study focused on predicting the environmental impact of data centers in the Electric Reliability Council of Texas by 2030 found that data centers could reduce their carbon footprint if they were operated differently. Essentially, the findings were that a data center in that particular grid "would need to be willing to ramp down about 13-15% of its capacity per year…to achieve such a goal [of decreased carbon emissions]" as well as operating more flexibly when grid prices are high (Rhodes, et al.). This is only the current use of energy the technology we have now requires. It is concerning that as it becomes more and more integrated into our lives in the future, the world will need to use even more energy. This is why studies like this, which give us actual ways we can reduce our carbon footprint and general impact on the environment, are so essential.

Policymakers must stay aware of the exponentially increasing amount of energy the world uses now, how that impacts our physical world, and what we can do to mitigate the effects so that we still have a physical world in the future.

The issue is not just that creating technology takes a near incomprehensible amount of energy; it is also that it needs to be maintained, developed, shipped around the world, and disposed of properly. Technology requires minerals, and mining for minerals uses up fossil fuels and water, subsequently releasing carbon dioxide and other greenhouse gases. Mineral mining is just the tip of the iceberg, though. It's not just data centers that use energy; in fact, just about all manufacturing efforts of technologies require huge amounts of energy. Shipping pieces and items around the world pollute the air, as well. (Okafor) The way we create technology is a relatively new concept. It is essential that we acknowledge the impact it has on our environment and stay vigilant of these facts.

Another incredibly important part of the conversation is the disposal of technology. In the UN Environment Programme from 2019, it was reported that $62.5 billion of electronic waste is thrown away, and only one-fifth of that is recycled formally (UN). And that number is only expected to increase, particularly as technology that hasn't been designed for longevity gets older. In fact, "experts forecast hundreds of thousands of tons of old wind turbine blades, batteries, and solar modules will need to be disposed of or recycled in the next decade—and millions of tons by 2050" (Scott). The wasteland of battered products that could be clogging up bodies of water, stunting natural growth, and spilling over into the population is a scary thought. Millions of tons in the next thirty years are a lot to fathom, particularly when we already have current landfill issues.

Unfortunately, this issue is not being addressed as completely as it should. There is a lot of deferment of responsibility regarding electronic disposal, whether it be from corporations classifying that as consumer responsibility or the federal government pushing it on to state responsibility. But unfortunately, "state laws often reference standards and international agreements forbidding the export of toxic e-waste to developing countries, for instance, but states lack jurisdiction over the matter" (Staff). Regardless of whether accountability is being tossed between individuals, corporations, state laws, or the federal government, the issue persists. It may be acceptable to pass some of the responsibility on to individuals and states when there is more public education and initiatives regarding how to handle old electronics, but governments are not emphasizing this currently. It is essential that governments prioritize both education of the public as well as finding a balance between individual and corporate responsibility for how our current consumption of technology impacts our environment.

Corporate responsibility is very likely going to be a buzzword going into the next half-century. The way most corporations are currently operating is not sustainable for the future. Fortunately, we are already seeing small-scale efforts at forcing corporations to be accountable for their products. One such point of contention where French courts recognized a conflict of interest between corporations and the greater good was the issue of planned obsolescence in Apple's iPhones. Apple unexpectedly ended up admitting to making iPhones that were not meant to last and apologized, though they deflected with an issue about software updates and battery capacity over time. (BBC) This is a direct violation of consumer trust, beyond being a truly alarming concept in terms of all the electronic waste that would be created by creating products meant to burn out quickly. Under French law, this is illegal. However, that is not the case everywhere. Many nations simply have little to no legislation regarding these issues because they are new and under the radar at the moment. Society gets caught up in using the devices, forgetting that every individual action has a greater impact.

Even just the face of business will surely be drastically different in the coming years as technology forces businesses and individuals to keep up or get left behind. It is likely that the monopolization of tech companies will get worse before it gets better. Rob Wilson, the founder of conversation artificial intelligence company OneReach.ai, seems to think that many companies will fail because switching to technological-based service from nothing “will require implementing a vast and exhaustive strategy for reorganizing internal technology and data, as well as a commitment to building automations right away. These early automations will fail, as will many of their lackluster successors” (OneReach). It will not be an easy transition, and that does ring alarm bells for small, locally owned businesses that may not have the automation capabilities of bigger firms. Imagine a future of fully automated chain restaurants, with none of the charisma of your local neighborhood deli. The world can only hope that not all its charms is lost to eventual total automation.

One of the biggest ways experts predict automation will change the future is by transforming the workforce with machine learning. Definitionally, machine learning is “an umbrella term that refers to a broad range of algorithms that perform intelligent predictions based on a data set” (Nichols, et al.). To put it simply, it describes the ongoing development of programming machines to mimic the human learning process and extrapolate information based on past experiences or knowledge. Machine learning is so valuable because it can potentially complete tasks much more efficiently than humans, especially jobs that involve repetitive or monotonous tasks because “robots are increasingly capable of helping lighten humans’ workload, don’t need lunch or rest breaks and are often capable of producing more predictable results with less downtime and error" (Steinberg). Of course, the concept of automation is not new and has been developing since Ford's creation of the assembly line. However, machine learning is much more than automation. Imagine a world where it is the norm to have machines serving your food or bagging your groceries or even fulfilling a managerial role at your work. Today, machines don't have the capabilities to interact with humans to that degree, but it does not seem so far off as they learn more and more about human culture.

This concept of machines replacing human jobs is a divisive one. Maybe it's driven by sci-fi horror genres, but people tend to imagine a world taken over by robots at the expense of humans. Luckily, this notion out there that machines will be stealing people's roles in society is extremely unlikely. The whole goal of artificial intelligence and automation is to have the ability to delegate tasks that take up inordinate human resources to machines so human attention can be directed elsewhere, not nowhere. Despite the controversy behind machines becoming further entrenched in society, it is happening under the world's nose. Per the World Economic Forum 2020 report, "AI will displace 85 million jobs but create 97 million new jobs across 26 countries by 2025" (Kande, et al.). That is only three years from now, and those numbers still hold relatively true. It doesn't take an overly imaginative individual to conceptualize how drastic of a shift in the workforce there will likely be in ten years, let alone fifty. The important part of the World Economic Forum's research is that despite the displacement of other jobs, more jobs are being created to develop and perfect the technology used in artificial intelligence. Granted, these may be higher-level jobs that require very advanced training, so some level of concern for lost lower-level jobs is justified. That being said, this could be positive for society if leadership prioritizes the technical education and training of citizens. Therefore, it is important for governments to be proactive about this shift in society, and to start having conversations on behalf of citizens about the future of the relationship between technology and humans.

Beyond corporate capability is the question of how to keep corporations that are capable of large-scale automation in check. What exactly is the responsibility of corporations as we move forward into an era of truly enmeshed technology? One recent example is the issue of the Algorithmic Accountability Act of 2019, which addressed the serious concern the public had over corporations using unlegislated artificial intelligence. Take the Amazon hiring artificial intelligence scandal, in which the algorithm decided that men were better suited for jobs because they represented a majority of the applications and developed a bias against women applicants. (Bradford) If Amazon had not needed to be transparent about this process, it may have gone on indefinitely and unsupervised. Artificial intelligence is not perfect, and though the way it develops is out of a company's hands, it is its job to ensure legitimacy on behalf of the public. This is yet another issue of corporate responsibility because corporations are in positions of power simply by operating this technology on the scale that they can.

Whether the upcoming concerns with technology are the redefinition of the workforce, shifting corporate responsibility, or its environmental impact, it is certain that technology will play a crucial role in the future. It is extremely important that it not be left out of discussions regarding economic and social policies in the future, lest we go past a point of no return.

Conclusion

The present moment is a turning point for society. The way that we as individuals act in the present is going to ultimately affect the future of our world. Earth cannot sustain itself with the amount of population growth and overcrowding that is currently occurring. This is causing a multitude of problems including natural resource scarcity, declining trends in conservation, and human impact on the environment. The role that corporations will play in our future has never been as impactful and transformative as it is today and will be in the coming years. As put most eloquently in an extremely comprehensive study titled *Technology, Megatrends, and Work: Thoughts on the Future of Business Ethics*: "In terms of ethics and morality, this means that the world's biggest corporations are not simply subject to extant moral concepts and norms, but agents that can transform or redirect them." (D'Cruz, et al.) It is up to us to ensure that corporations and governments work in tandem with technology and sustainability factors as agents of good, aware of their own social responsibility to both consumers and our environment, instead of entities profiting from the degradation of our future. There will be no benefits if we continue to act selfishly.

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